Research progress: factor measurement and forecast sharing

November 3, 2026

Research question and initial plan

Industries can measure common conditions (M) or help estimate predictive coefficients (O). We ask when assigning these roles separately improves risk forecasts. We hypothesize measurement expansion helps at fixed dimension, and sharing helps with short histories and similar slopes; heterogeneity, noise, or instability can offset gains. Motivated by principal-component forecasting [1] and panel-composition evidence [2], we study finite-sample variance/information tradeoffs. Dimension reduction cannot improve the unrestricted population optimum with identical information.

Completed foundation and first two experiments

We archived French 48-industry daily returns, both weightings, provenance, missingness, and hashes. Weekly $Q_{jt} = \sum_d r_{jd}^2$ and $v_{jt} = \log(Q_{jt} + 10^{-8})$ require at least three daily returns. These proxy daily-return variation. Own controls are current v and complete trailing four-/twelve-week means; the response is next calendar week's v.

Experiment 1 fixed Food, Mach, Banks, Oil, Util, and Hlth as targets and the small donor panel. Eighteen procedures compared panels, sharing, direct methods, and baselines. Selection preceded 2022-2024 evaluation. The selected all-48/common-slope factor gained **4.27%** against own history; equal-weighted gain was **1.66%**, including **2.95% worse Banks MSE**. Broader measurement supplied most of the gain.

Experiment 2 added paired stationary bootstraps [3] and coefficient histories 104/260/520/expanding, with expanding PCA. In 2025-August 2026, no history beat own history or ridge after Holm-19 adjustment. L=104 beat elastic net and PLS under the approximate rule. No finite history showed an adjusted advantage over expanding coefficients.

Experiment 3: completed November 3

We mapped **K=1-8 factors** against **L=52,78,. . . ,520 coefficient pairs**, plus expanding controls, for separate/common/partial slopes. Part A fitted PCA/scalers and coefficients on the same recent L rows; Part B used expanding PCA with identical coefficient windows. Measurement and common/partial outcome sharing used all 48 industries. Own coefficients stayed industry-specific; partial-pooling penalties used historical target validation.

All configurations shared 718 earlier dates (2011-2024) and 87 later dates (2025-August 2026), both already inspected. Results are exploratory; forecast windows, tuning, losses, and inference were independently audited.

For value-weighted portfolios, the table fixes K=2/common slopes. Gains average six targets' percentage MSE reductions against their matching-history own regressions.

Coefficient pairs	Earlier: matched PCA	Later: matched PCA	Later: expanding PCA
52	5.73%	1.66%	1.96%
104	4.15%	3.53%	3.97%
260	5.47%	-2.00%	-1.75%
520	5.08%	-0.75%	-0.41%

History effects depend on period and weighting. Later value gains favored roughly 78-104 pairs over 260-520, without falling monotonically. Earlier gains persisted with long histories. Later equal-weighted K=2/common gained 0.92% at matched L=104 and 3.01% with expanding coefficients/PCA. A universal short-history advantage is unsupported.

Relative edge differs from absolute accuracy. Later value-weighted K=2/common MSE was 0.7516 at matched L=104, versus 0.7181 with expanding coefficients/PCA, although gains were 3.53% versus 1.23%. The longer-history own-regression benchmark was also stronger.

Pooling cushioned extra-factor costs. At L=52, later value/matched-PCA gains changed from K=1 to K=8: separate +2.93% to -11.24%, common +3.40% to +1.37%, partial +3.29% to -0.75%. Longer history helped separate slopes catch up earlier. This is consistent with lower estimation variance from sharing. Partial pooling did not reliably improve on common slopes.

Longer PCA history was not uniformly beneficial. Expanding PCA helped the K=2 value anchors, but reduced gains by a median 1.2-1.6 percentage points across finite histories in the later value grid. It changed scaling/loadings, downstream fits, and partial-pooling penalty choices. Neither this contrast nor the pooling pattern identifies a causal mechanism. All unfavorable configurations remain reported.

Bootstraps used 9,999 draws of saved losses, without refitting/retuning. **None of 30 later-period anchors passed primary block-13 Holm adjustment in either weighting.** Later value/matched K=2/common/L=104 gained 3.53%, with nominal 95% marginal interval [-3.18%, 12.18%]. Two equal-weighted block-26 sensitivity flags did not replace the primary result.

Inference limits and pending work

Bootstrap calibration remains unresolved. The inherited 100-replication diagnostic rejected a nominal 5% null 11% of the time at 87 dates; nominal 95% coverage was 90% at 87 dates and 81% at 156. Holm cannot repair marginal miscalibration. Audits verify computations, not coverage. Dependence, nested models, fixed targets, source revisions, and short evaluation histories constrain claims; non-rejection does not establish equivalence.

Next steps and questions for discussion

Before another empirical grid, we propose agreeing on a narrow hypothesis and confirmation design: does pooling help most when coefficient history is short relative to model dimension? Experiment 3 fixed all 48 measurement industries; choosing different measurement and coefficient-sharing donor sets remains open.

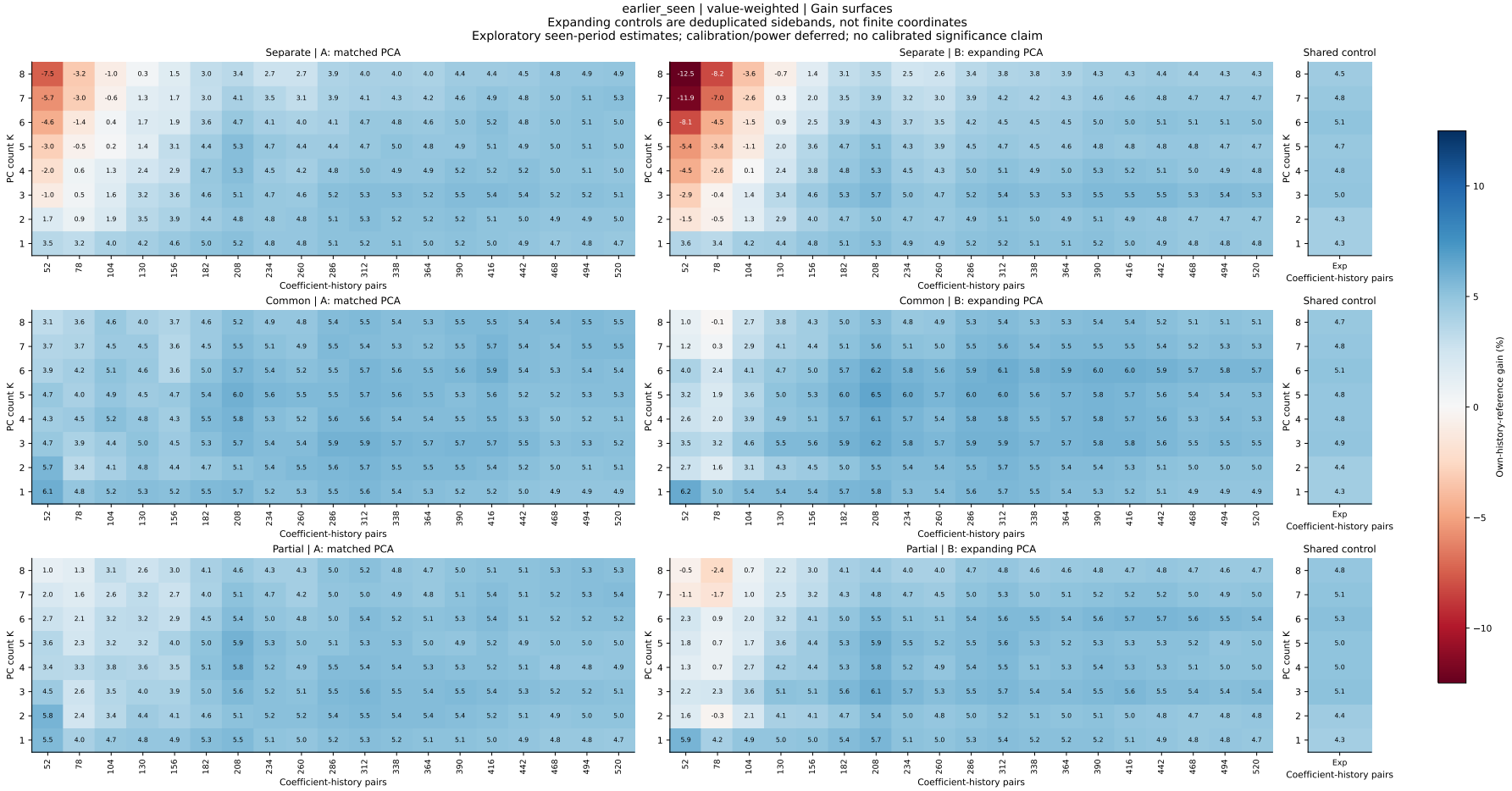
Calibration and power studies remain **deferred and pending**; their design is a priority for discussion. We propose comparing false-rejection rates, coverage, and power under controlled dependence and known forecast advantages, including repeated fitting/tuning. Measurement-noise/slope-heterogeneity experiments and learned donor selection remain later work. Empirical confirmation requires an uninspected period or market.

Protocols, complete results, graphs, and audits: Experiment 1, Experiment 2, Experiment 3. The original proposal remains archived.

Selected references

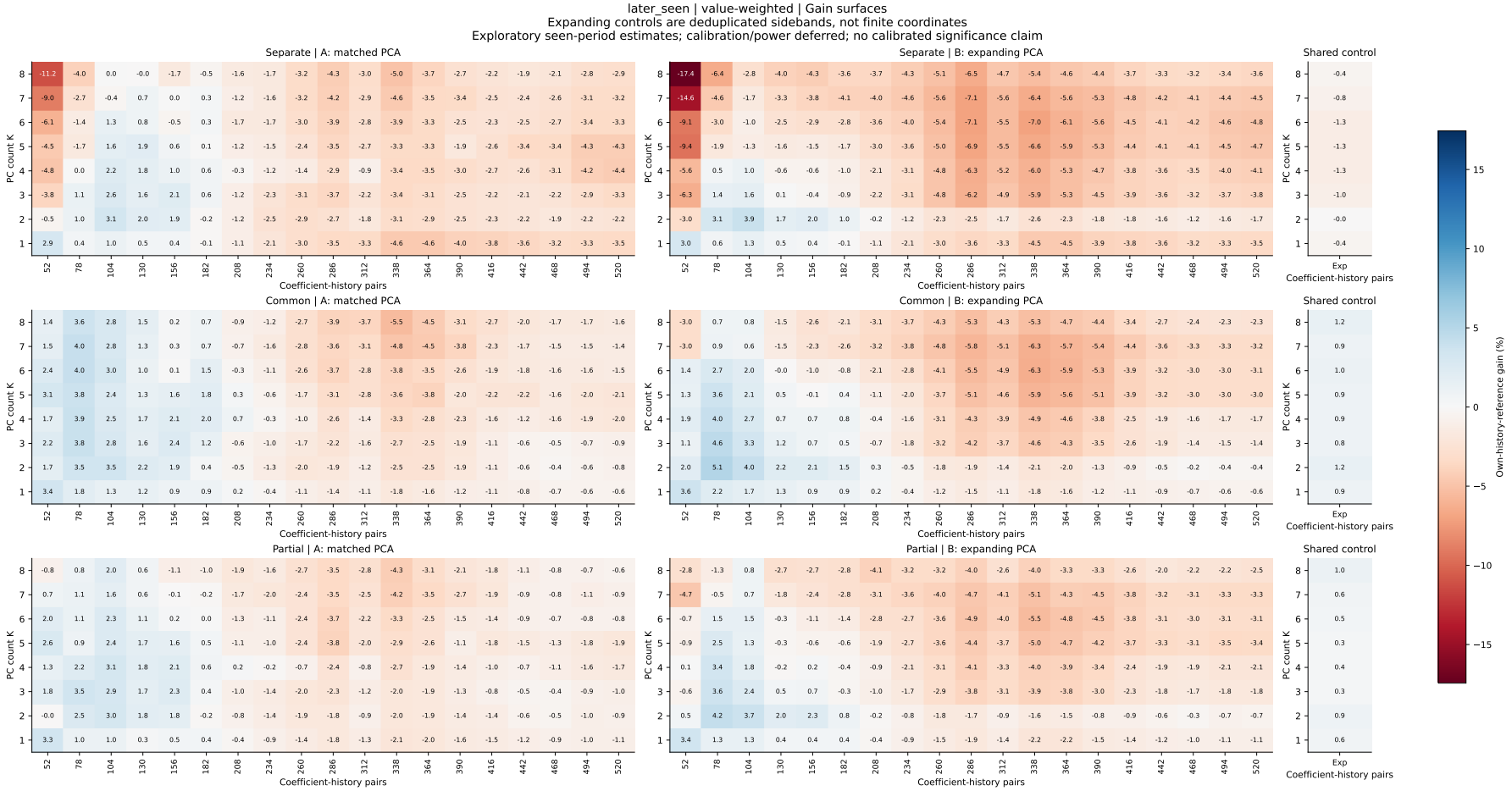
1. Stock and Watson (2002), *Forecasting Using Principal Components from a Large Number of Predictors.* DOI.
2. Boivin and Ng (2006), *Are More Data Always Better for Factor Analysis?* DOI.
3. Politis and Romano (1994), *The Stationary Bootstrap.* DOI.

Figure 1: earlier-period forecasting gains



2011-2024, 718 forecast dates; exploratory, previously inspected. Horizontal axes show coefficient-history pairs L; vertical axes show factors K. Rows compare separate/common/partial slopes; columns compare matched/expanding PCA. Blue indicates lower MSE than matching-history own regression, averaged equally across six targets. Expanding-coefficient controls appear separately at right. Longer histories retain broadly favorable gains; sharing cushions high-K/short-history deterioration. Colors show point estimates, not significance.

Figure 2: later-period forecasting gains



2025-August 2026, 87 forecast dates; exploratory, previously inspected. Axes, rules, and reference definitions match Figure 1. Short windows often show a larger relative edge, while additional factors particularly hurt separate slopes. Partial pooling and expanding PCA are not uniformly superior. The changing own-history benchmark means these maps do not rank absolute accuracy. Color limits differ across periods: compare cell values, not shades. No primary Holm-adjusted anchor rejects.